LOW-LIGHT IMAGE ENHANCEMENT
& OBJECT DETECTION

Zero-DCE + YOLOv8 Pipeline

Technical Stack Report

COURSE

AI335L — Deep Learning

PHASE

Phase 2 of 6 — Foundations: Data, EDA and Baseline Model

REPOSITORY

github.com/Nimra-waseemaftab/image_enhancement_and_detection

TEAM MEMBERS

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1. LIBRARIES AND TOOLS

The project is implemented in Python using PyTorch as the core framework. All phases rely on the libraries listed below.

Library / Tool	Role in Project
PyTorch	Core deep learning framework; dynamic computation graph, GPU tensor operations
torchvision	Pretrained models (MobileNetV2), image transforms, augmentation utilities
Ultralytics	YOLOv8 API; single-stage real-time object detection on enhanced images
NumPy	Numerical array operations, brightness statistics, IQR outlier detection
Pillow (PIL)	Image loading, resizing, format conversion across all phases
matplotlib	EDA charts, loss curves, feature map and Grad-CAM visualisations
scikit-image	PSNR and SSIM computation for quantitative image quality evaluation
OpenCV (cv2)	Grad-CAM heatmap colouring, colour space ops, image overlays
Optuna	Bayesian HPO (TPE sampler, 25 trials, SQLite persistence)
tkinter	Native OS GUI for the NightVision interactive inference application
ipywidgets	Notebook-embedded file upload widget for quick in-notebook inference demos
csv / json	Lightweight experiment tracking (MetricWriter); config serialisation

2. DATA AND PREPROCESSING TECHNIQUES

The LOL-v2 dataset provides paired low/normal images across Synthetic (~689 pairs) and Real-Captured (~100 pairs) subsets. All preprocessing decisions were guided by EDA findings.

Technique	Description	Phases
Dataset (LOL-v2)	Synthetic + Real-Captured subsets; low-light inputs only used for Zero-DCE; paired for supervised evaluation	All
Resize 256x256	Uniform resolution; Zero-DCE is fully-convolutional so inference size remains flexible	2-5
Normalize [0, 1]	ToTensor() converts uint8 to float32; no ImageNet mean subtraction (unsupervised task)	3-5
Train/Val Split	90/10 random split; fixed seed=42; split before dataset construction to prevent leakage	3-4
Augmentation	Training only: horizontal flip, rotation +/-15 deg, ColorJitter, RandomResizedCrop(0.8-1.0)	3
Corruption Check	PIL verify() scan; zero-byte and unreadable images removed before any training	2

Technique	Description	Phases
Outlier Detection	IQR-based brightness outlier flagging (1.5xIQR); images inspected before exclusion	2
Paired Dataset	Low and normal images matched by filename for supervised transfer learning training	4-5

3. MODEL AND TRAINING TECHNIQUES

Three model architectures are trained and compared. Training infrastructure includes mixed-precision, cosine scheduling, gradient clipping, early stopping, and Bayesian HPO.

Model / Technique	Key Details	Phase
MLP Baseline	3-layer MLP (3-64-64-3, BatchNorm, ReLU, Dropout); pixel-wise gamma pseudo-target; MSE loss	3
DCENet (Zero-DCE)	7-layer CNN, U-Net skip connections; predicts 24-ch curve maps (8 iter x 3 RGB); ~79K params; no paired supervision	3-4
DCENetPro	Enhanced DCENet with MaxPool2d, BatchNorm2d, Dropout2d, bilinear decoder; same 24-ch output	4
Transfer Learning	MobileNetV2 (ImageNet) encoder + PixelShuffle decoder; Feature Extraction / Full Fine-Tune / Differential LR configs	4
VAE	Convolutional VAE on low-light images; samples synthetic augmentations; KL divergence monitored for collapse	4
Adam + Weight Decay	lr=1e-4, weight_decay=1e-4; differential LRs in TL (head: 1e-3, backbone: 1e-5)	3-4
CosineAnnealingLR	Smooth LR decay to 1% of initial over total epochs; prevents sharp drops	4
Gradient Clipping	max_norm=1.0; prevents exploding gradients in early training	3-4
Mixed Precision (AMP)	GradScaler + autocast; halves VRAM usage, accelerates CUDA training	4
Early Stopping	Patience=20 on validation loss; best checkpoint restored on trigger	3-4
Unsupervised Loss	4 physics losses: Illumination Smoothness (w=200), Spatial Consistency (w=1), Colour Constancy (w=5), Exposure (w=10)	3-4
Ablation Study	Run1: no regularisation. Run2: Dropout+Augmentation+WD. Run3: BN+Dropout+WD (DCENetPro)	4
HPO (Optuna)	Bayesian TPE; 25 trials; SQLite persistence; searches LR, dropout, weight_decay, loss weights	4

4. EVALUATION AND ACCURACY METRICS

Metric / Technique	Description	Phases
PSNR	Peak Signal-to-Noise Ratio (dB); primary quality metric; higher is better	3-5
SSIM	Structural Similarity Index [0,1]; evaluates luminance, contrast, and structure	3-5
MSE Loss	Mean Squared Error; training signal for MLP Baseline	3
L1 Loss (MAE)	Transfer learning loss; more robust to pixel outliers than MSE	4

Metric / Technique	Description	Phases
3-Seed Evaluation	Each model evaluated at seeds 42, 123, 7; mean +/- std PSNR/SSIM reported	4
Feature Map Vis.	Forward hooks at conv1/conv4; 16-channel activation grids (inferno colourmap)	4
Grad-CAM / Saliency	Gradient of mean enhanced brightness w.r.t. conv4 features; heatmap overlaid on input	4
Loss Curve Analysis	Per-epoch train/val loss logged; divergence indicates overfitting	3-4

5. DEPLOYMENT AND INTERFACE TOOLS

Phase 5 wraps the trained models into a notebook widget and a native GUI application for interactive end-user demonstration.

Tool / Platform	Role	Phase
Jupyter Notebook	Primary development environment; all phases self-contained in labelled cells	All
ipywidgets	FileUpload widget in Jupyter; runs full pipeline inline and shows side-by-side output	5
tkinter (NightVision)	Native OS GUI; neon-styled dark UI; file picker, pipeline controls, save button	5
YOLOv8 (Ultralytics)	Loaded via YOLO('yolov8x.pt'); runs at conf=0.15 on enhanced image; draws bounding boxes	5
torch.no_grad()	Disables gradient tracking at inference; reduces memory usage and speeds forward pass	5
model.eval()	Switches BatchNorm to inference stats; disables Dropout for deterministic output	5
PIL / ImageTk	Converts NumPy arrays to Tk-compatible images for the GUI canvas display	5
threading	Background thread prevents the Tkinter window from freezing during processing	5

6. FINAL SYSTEM OUTPUTS

Output / Component	Description	Phase
best_zero_dce_lolv2.pth	DCENet model weights at best validation loss checkpoint	4
best_mlp_baseline.pth	MLP Baseline weights for ablation and comparison	3
ablation_run1/2/3.pth	Three DCENet variant checkpoints from the ablation study	4
Enhanced Images	Low-light inputs processed through 8-iteration curve enhancement; output in [0,1]	5
YOLO Detection Output	Bounding boxes and class labels drawn on enhanced frames at conf=0.15	5
EDA Charts	Brightness histograms, RGB channel plots, resolution scatter, mean image heatmap	2
Training Loss Curves	Per-epoch total and component losses across all three ablation runs	3-4
PSNR/SSIM Comparison Table	Three-model comparison (MLP, DCENet, DCENetPro) with mean+/-std across 3 seeds	4

Output / Component	Description	Phase
Feature Maps	16-channel activation grids at conv1 and conv4 (inferno colourmap)	4
Grad-CAM Overlay	Saliency and Grad-CAM heatmaps overlaid on test images	4
Optuna Study Results	25-trial HPO results with best hyperparameter set and trial charts	4
metrics.csv	Per-epoch scalar log (loss, LR, PSNR, SSIM) written by MetricWriter	3-4
NightVision GUI	Standalone Tkinter app showing original, enhanced, and detected images with save	5